

```
graph TD; X["Input Matrix X  
• Type: numpy.ndarray (float64)  
• Dimensions:  $N \times n$  (C-contiguous)  
• Footprint:  $N \cdot n \cdot 8$  bytes  
• Resident: 400 MB ( $N = 100k$ )"] --> F["extract_features(X)"]; F --> Y["Output Matrix Y  
• Type: numpy.ndarray / DataFrame  
• Dimensions:  $N \times 33$  features  
• Footprint:  $N \cdot 33 \cdot 8$  bytes  
• Allocated: 25.2 MiB ( $N = 100k$ )  
• Direct zero-copy return"]
```

User Application Space (Python)

Input Matrix X

- Type: numpy.ndarray (float64)
- Dimensions: $N \times n$ (C-contiguous)
- Footprint: $N \cdot n \cdot 8$ bytes
- Resident: 400 MB ($N = 100k$)

↓

extract_features(X)

Output Matrix Y

- Type: numpy.ndarray / DataFrame
- Dimensions: $N \times 33$ features
- Footprint: $N \cdot 33 \cdot 8$ bytes
- Allocated: 25.2 MiB ($N = 100k$)
- Direct zero-copy return

Native Computational Core (Rust Extension: `_core.pyd` / `_core.so`)

PyO3 FFI Ingestion

- PyReadOnlyArray2 view
- $O(1)$ Contiguity check
- Borrow unowned slice: `&[f64]` of len $N \cdot n$
- Zero copy (+0.0 MB)
- Detach CPython GIL: `py.detach(|| ...)`

Rayon Work-Stealing

- Thread pool ($p = 16$ threads)
- Split along Series Axis (N): $\mathbf{x}_i \in \mathbb{R}^n$ (independent)
- Lock-free concurrency
- Thread-local scratchpads:
 - `ORDER_BUF` (selection)
 - `WORKSPACE` (RealFFT)
- Aux mem: $O(p \cdot n) \approx 0.1$ MB

Fused Kernel ($\mathbf{x} \in \mathbb{R}^n$, Worker Thread)

Pass 1: Accumulations
sum, sum^2 , min, max, short-circuit NaN check

Pass 2: Central Moments
mean μ , variance σ^2 , std σ , skewness S , kurtosis K

Pass 3: Successive Diffs
mean abs change, central 2nd deriv, CID_CE

Pass 4: Runs & Crossings
zero & mean crossings, peak count ($s=3$), run strikes

Pass 5: Autocorr & Trend
autocorr lags $\{1, 2, 5, 10\}$, linear trend slope β , r^2

Specialized Algorithmic Kernels
Quickselect ($O(n)$) · RealFFT (Hermitian, $O(n \log n)$)
Permutation Entropy (branchless 3-bit LUT)

Direct Parallel Write of $N \times 33$ Features
(Zero Serialization Overhead)

Incremental Streaming Engine (StreamingExtractor):

- `push(val)`: Strictly $O(1)$ sample updates for running moments ($S_1 \dots S_4$), trend covariance, diffs, zero crossings.
- `compute_features()`: On-demand snapshot evaluation (Quickselect, RealFFT) over circular ring buffer (capacity W).

Input Matrix X

- Type: `numpy.ndarray (float64)`
- Dimensions: $N \times n$ (C-contiguous)
- Footprint: $N \cdot n \cdot 8$ bytes
- Resident: 400 MB ($N = 100k$)

- ## Input Matrix X
- Type: `numpy.ndarray (float64)`
 - Dimensions: $N \times n$ (C-contiguous)
 - Footprint: $N \cdot n \cdot 8$ bytes
 - Resident: 400 MB ($N = 100k$)

```
extract_features(X)
```

Output Matrix Y

- Type: `numpy.ndarray` / `DataFrame`
- Dimensions: $N \times 33$ features
- Footprint: $N \cdot 33 \cdot 8$ bytes
- Allocated: 25.2 MiB ($N = 100k$)
- Direct zero-copy return

- ## Output Matrix Y
- Type: `numpy.ndarray` / `DataFrame`
 - Dimensions: $N \times 33$ features
 - Footprint: $N \cdot 33 \cdot 8$ bytes
 - Allocated: 25.2 MiB ($N = 100k$)
 - Direct zero-copy return

PyO3 FFI Ingestion

- PyReadonlyArray2 view
- $O(1)$ Contiguity check
- Borrow unowned slice:
 $\&[f64]$ of len $N \cdot n$
- Zero copy (+0.0 MB)
- Detach CPython GIL:
 `py.detach(|| ...)`

- ## PyO3 FFI Ingestion
- PyReadonlyArray2 view
 - $O(1)$ Contiguity check
 - Borrow unowned slice:
 $\&[f64]$ of len $N \cdot n$
 - Zero copy (+0.0 MB)
 - Detach CPython GIL:
 `py.detach(|| ...)`

&[f64] of len $N \cdot n$

Rayon Work-Stealing

- Thread pool ($p = 16$ threads)
- Split along Series Axis (N):
 $\mathbf{x}_i \in \mathbb{R}^n$ (independent)
- Lock-free concurrency
- Thread-local scratchpads:
 - ORDER_BUF (selection)
 - WORKSPACE (RealFFT)
- Aux mem: $O(p \cdot n) \approx 0.1$ MB

- ## Rayon Work-Stealing
- Thread pool ($p = 16$ threads)
 - Split along Series Axis (N):
 $\mathbf{x}_i \in \mathbb{R}^n$ (independent)
 - Lock-free concurrency
 - Thread-local scratchpads:
 - ORDER_BUF (selection)
 - WORKSPACE (RealFFT)
 - Aux mem: $O(p \cdot n) \approx 0.1$ MB

- Lock-free concurrency

```

graph TD
    subgraph Fused_Kernel [Fused Kernel  $(x \in \mathbb{R}^n, \text{Worker Thread})$ ]
        direction TB
        P1[Pass 1: Accumulations  
sum,  $\text{sum}^2$ , min, max, short-circuit NaN check]
        P2[Pass 2: Central Moments  
mean  $\mu$ , variance  $\sigma^2$ , std  $\sigma$ , skewness  $S$ , kurtosis  $K$ ]
        P3[Pass 3: Successive Diffs  
mean abs change, central 2nd deriv,  $CID\_CE$ ]
        P4[Pass 4: Runs & Crossings  
zero & mean crossings, peak count (s=3), run strikes]
        P5[Pass 5: Autocorr & Trend  
autocorr lags {1, 2, 5, 10}, linear trend slope  $\beta$ ,  $r^2$ ]
    end

    subgraph Specialized_Algorithmic_Kernels [Specialized Algorithmic Kernels]
        direction TB
        Q[Quickselect  $(O(n))$ ]
        R[RealFFT (Hermitian,  $O(n \log n)$ )]
        P[Permutation Entropy (branchless 3-bit LUT)]
    end
  
```

Fused Kernel ($x \in \mathbb{R}^n$, Worker Thread)

- Pass 1: Accumulations**
sum, sum^2 , min, max, short-circuit NaN check
- Pass 2: Central Moments**
mean μ , variance σ^2 , std σ , skewness S , kurtosis K
- Pass 3: Successive Diffs**
mean abs change, central 2nd deriv, CID_CE
- Pass 4: Runs & Crossings**
zero & mean crossings, peak count (s=3), run strikes
- Pass 5: Autocorr & Trend**
autocorr lags {1, 2, 5, 10}, linear trend slope β , r^2

Specialized Algorithmic Kernels

- Quickselect ($O(n)$)
- RealFFT (Hermitian, $O(n \log n)$)
- Permutation Entropy (branchless 3-bit LUT)

Pass 1: Accumulations
sum, sum², min, max, short-circuit NaN check

Pass 2: Central Moments mean μ , variance σ^2 , std σ , skewness S , kurtosis K

Pass 3: Successive Diffs mean abs change, central 2nd deriv, <i>CID_CE</i>
--

Pass 4: Runs & Crossings
zero & mean crossings, peak count ($s=3$), run strikes

Pass 5: Autocorr & Trend
autocorr lags {1, 2, 5, 10}, linear trend slope β , r^2

Specialized Algorithmic Kernels
 Quickselect ($O(n)$) · RealFFT (Hermitian, $O(n \log n)$)
 Permutation Entropy (branchless 3-bit LUT)

Specialized Algorithmic Kernels
 Quickselect ($O(n)$) · RealFFT (Hermitian, $O(n \log n)$)
 Permutation Entropy (branchless 3-bit LUT)

Direct Parallel Write of $N \times 33$ Features (Zero Serialization Overhead)

Incremental Streaming Engine (StreamingExtractor):

- `push(val)`: Strictly $O(1)$ sample updates for running moments ($S_1 \dots S_4$), trend covariance, diffs, zero crossings.
- `compute_features()`: On-demand snapshot evaluation (Quickselect, RealFFT) over circular ring buffer (capacity W).

- Incremental Streaming Engine (StreamingExtractor):**
- `push(val)`: Strictly $O(1)$ sample updates for running moments ($S_1 \dots S_4$), trend covariance, diffs, zero crossings.
 - `compute_features()`: On-demand snapshot evaluation (Quickselect, RealFFT) over circular ring buffer (capacity W).